Why Do Employees Leave? A Beginner's Journey Through the IBM HR Attrition Data

I am a developer and a beginner in data analysis. I had just finished an Excel course, and I wanted to learn by doing something real. So I picked one question:

Which things are connected to employees leaving a company?

This is the story of how I tried to answer it: what I found, what confused me, and one mistake that taught me more than the results.

> **Please note:** the IBM HR dataset is fictional, created by IBM data scientists. What I found describes this dataset only, and everything here is an **association, not a cause**.

Chapter 1: Meeting the data

The dataset has 1,470 employees. 237 of them left, which is 16.1%. I sorted the variables into three types, because each type needs different tools: *scale* (numbers like income and years), *ordinal* (ordered levels like satisfaction), and *nominal* (labels like job role). I removed three columns that never change (EmployeeCount, StandardHours, Over18), since they cannot explain anything.

Then I started in Excel, where I felt comfortable, and asked simple questions.

Does overtime matter? It was the first surprise. Among people who work overtime, **30.5% left**. Among people who don't, **10.4%**.

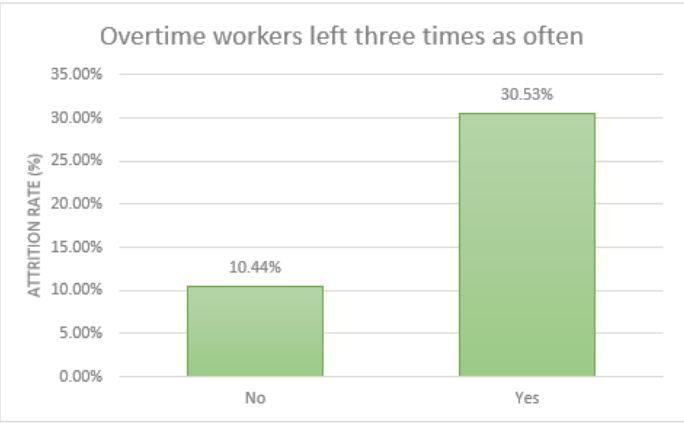


***Figure 1**: Attrition rate by overtime status*

Does seniority matter? Entry-level employees left the most (**26.3%**). At the top levels the rate was only 4.7% and 7.3%.

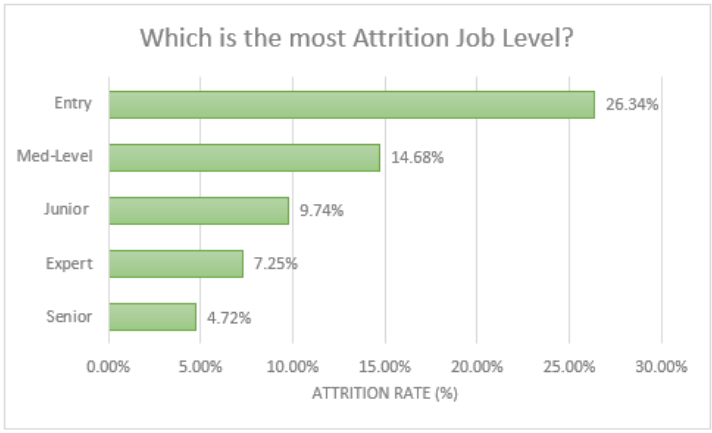


***Figure 2:** Attrition rate by job level*

Which roles are at risk? Sales Representatives stood out at about 40%, with Laboratory Technicians and Human Resources next. In the chart, bars show how many people are in each role and the line shows the attrition rate.

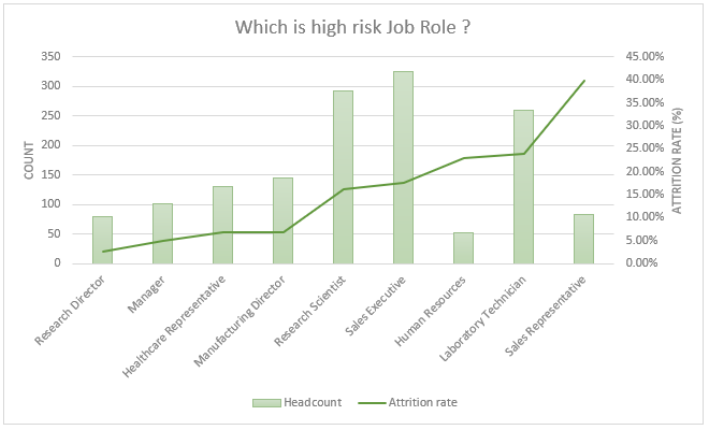


***Figure 3**: Headcount (bars) and attrition rate (line) by job role*

Does travel matter? Yes: 8.0% for non-travellers, 15.0% for people who travel rarely, and 24.9% for people who travel frequently.

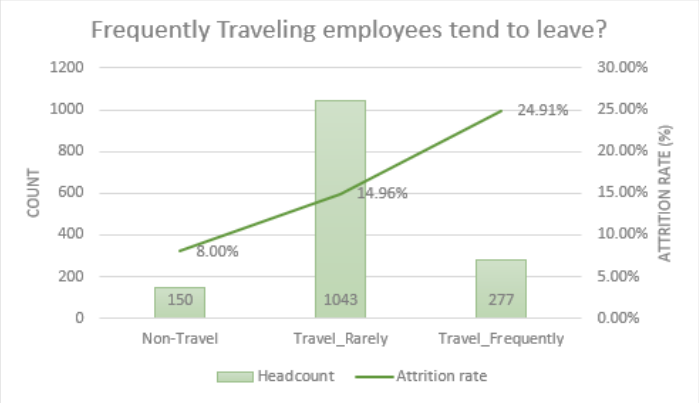


Figure 4: *Headcount (bars) and attrition rate (line) by business travel*

Does satisfaction matter? Attrition was 22.8% at the lowest job satisfaction and 11.3% at the highest. But the middle levels were almost the same (about 16.5%), so it was not a perfectly steady slope. I remembered this later.

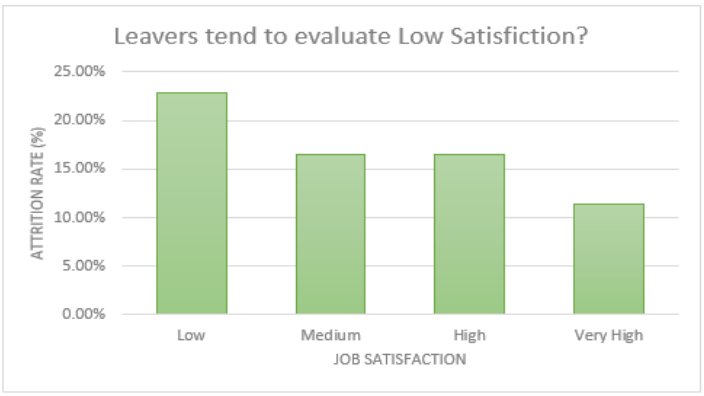


Figure 5: *Attrition rate by job satisfaction level*

I also looked at age, department and salary. Younger employees left more often (the youngest group has only 28 people, so its 57% is not reliable), and Sales and HR had higher rates than Research & Development. When I compared the salaries of leavers and stayers, they were similar within each job level, which told me that pay mostly follows job level.

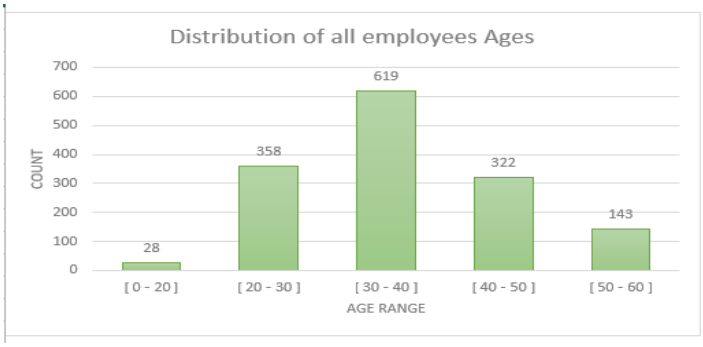


Figure 6: *Age distribution of all employees*

The distribution of leavers employees per age is shown below

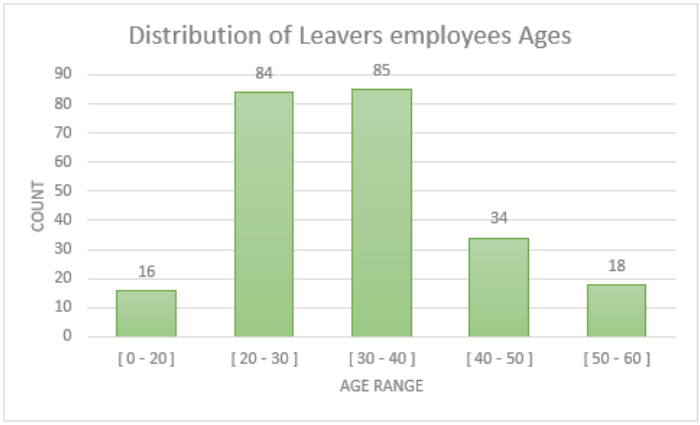


Figure 7: *Age distribution of employees who left*

Attrition rate for each age range

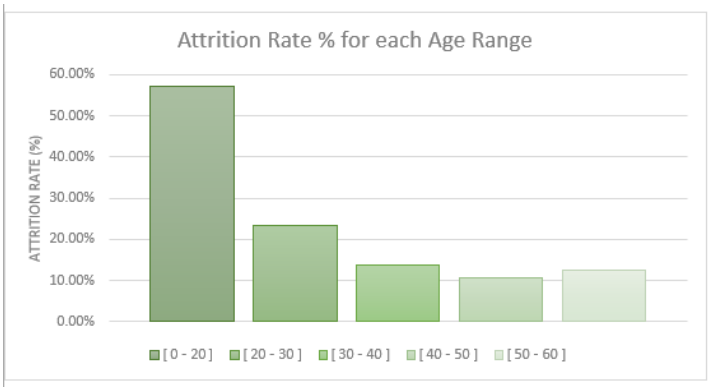


Figure 8: *Attrition rate by age range*

Next, we'll look at the distribution of older vs. younger generations within organizations and compare the two groups' ages to see how that distribution appears

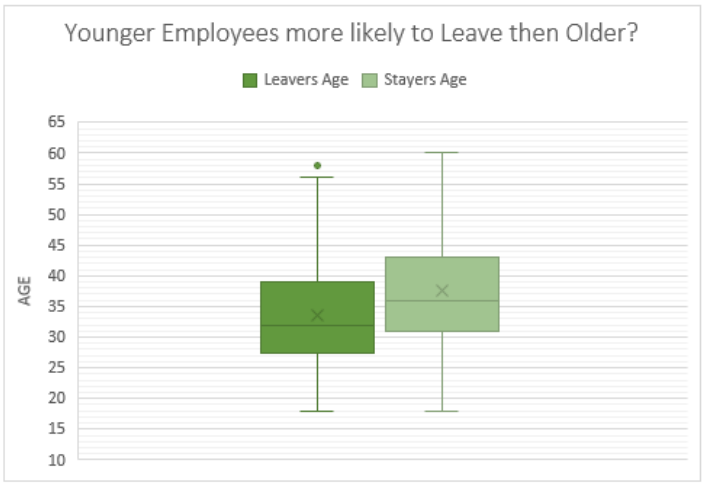


Figure 9: *Age box plot: leavers vs stayers*

Then, we look at top risk department by compare and attrition rate of each department with the headcount, which was **Human Resources and Sales.**

Attrition rate for each department

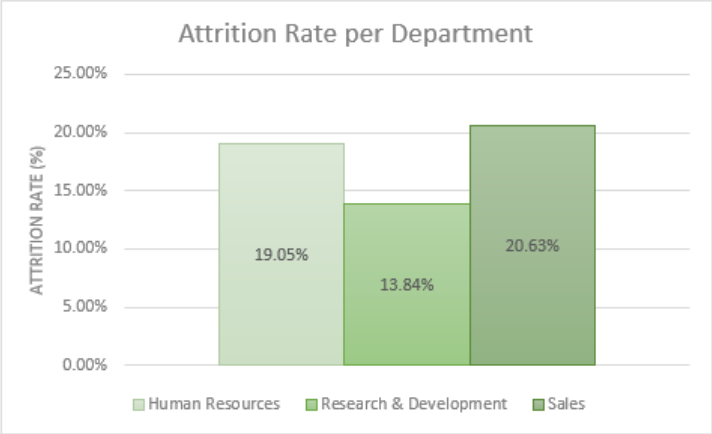


Figure 10: *Attrition rate by department*

The distribution of employees per department is shown below.

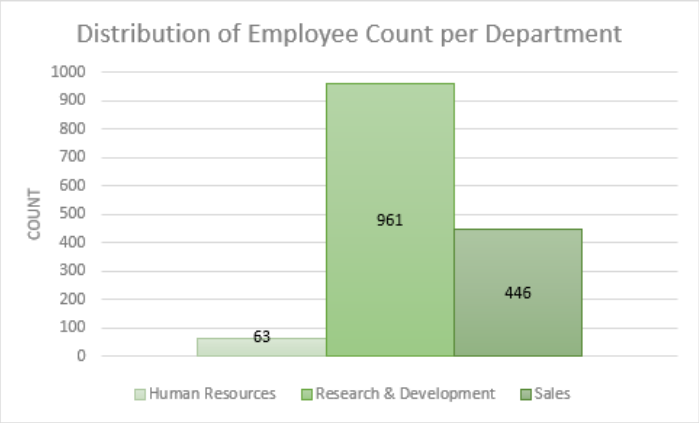


Figure 11*: Headcount by department*

Compared the salaries of leavers and stayers, for each job level and department

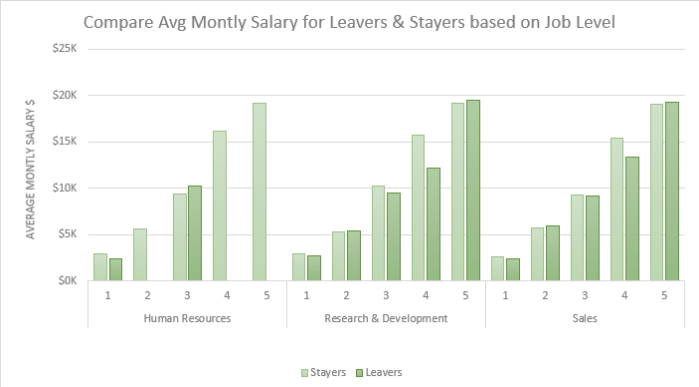


Figure 12: *Average monthly salary by department and job level, stayers vs leavers*

Chapter 2: Is it real, or just chance?

Charts show patterns, but patterns can be luck. I moved to SPSS and tested each variable against Attrition. The important lesson: the right test depends on the type of data, and a tiny p-value is not enough, because with 1,470 rows almost everything is "significant". So I always looked at the **effect size** too.

Question	Test	Result
Job role and attrition	Chi-square	$p < .001$, Cramér's V = 0.242 (small to medium)
Overtime and attrition	Chi-square	$\chi^2(1) = 89.04$, $p < .001$, Cramér's V = 0.246 (small to medium)
Business travel and attrition	Chi-square	$\chi^2(2) = 24.18$, $p < .001$, Cramér's V = 0.128 (small)
Department and attrition	Chi-square	$\chi^2(2) = 10.80$, $p = .005$, Cramér's V = 0.086 (very small)
Job level and attrition	Mann-Whitney U	$Z = -7.30$, $r \approx 0.19$: leavers sit at lower levels
Job involvement	Mann-Whitney U	$Z = -4.58$, $r \approx 0.12$ (small)
Job satisfaction	Mann-Whitney U	$Z = -3.95$, $r \approx 0.10$ (small)

Overtime turned out to have the largest effect size (V = 0.246), almost the same as job role (0.242), while department was the weakest (V = 0.086). Here is the SPSS output for the ordinal variables and for the other three crosstabs:

Ranks

	Attr...	N	Mean Rank	Sum of Ranks
JobLevel	No	1233	769.06	948252.50
	Yes	237	560.90	132932.50
	Total	1470		

Test Statistics[a]

	JobLevel
Mann-Whitney U	1.047E5
Wilcoxon W	1.329E5
Z	-7.296
Asymp. Sig. (2-tailed)	.000

a. Grouping Variable: Attrition

Figure 13: *SPSS Mann-Whitney U: job level by attrition*

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
JobRole * Attrition	1470	100.0%	0	.0%	1470	100.0%

JobRole * Attrition Crosstabulation

Count

		Attrition		
		No	Yes	Total
JobRole	Healthcare Representative	122	9	131
	Human Resources	40	12	52
	Laboratory Technician	197	62	259
	Manager	97	5	102
	Manufacturing Director	135	10	145
	Research Director	78	2	80
	Research Scientist	245	47	292
	Sales Executive	269	57	326
	Sales Representative	50	33	83
Total		1233	237	1470

Chi-Square Tests

	Value	df	Asymp. Sig. (2-sided)
Pearson Chi-Square	86.190[a]	8	.000
Likelihood Ratio	88.909	8	.000
Linear-by-Linear Association	6.624	1	.010
N of Valid Cases	1470		

a. 0 cells (.0%) have expected count less than 5. The minimum expected count is 8.38.

Symmetric Measures

		Value	Approx. Sig.
Nominal by Nominal	Phi	.242	.000
	Cramer's V	.242	.000
N of Valid Cases		1470	

Figure 14: *SPSS crosstab: job role by attrition (chi-square & Cramér's V)*

Mann-Whitney

Ranks

	Attr...	N	Mean Rank	Sum of Ranks
JobSatisfaction	No	1233	753.95	929619.50
	Yes	237	639.52	151565.50
	Total	1470		
JobInvolvement	No	1233	755.09	931025.00
	Yes	237	633.59	150160.00
	Total	1470		

Test Statistics[a]

	Job Satisfaction	Job Involvement
Mann-Whitney U	123362.500	121957.000
Wilcoxon W	151565.500	150160.000
Z	-3.946	-4.580
Asymp. Sig. (2-tailed)	.000	.000

a. Grouping Variable: Attrition

Figure 15: *Mann-Whitney U: job satisfaction and job involvement by attrition*

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
OverTime * Attrition	1470	100.0%	0	.0%	1470	100.0%

OverTime * Attrition Crosstabulation

			Attrition		
			No	Yes	Total
OverTime	No	Count	944	110	1054
		Expected Count	884.1	169.9	1054.0
	Yes	Count	289	127	416
		Expected Count	348.9	67.1	416.0
Total		Count	1233	237	1470
		Expected Count	1233.0	237.0	1470.0

Chi-Square Tests

	Value	df	Asymp. Sig. (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	89.044[a]	1	.000		
Continuity Correction[b]	87.564	1	.000		
Likelihood Ratio	81.402	1	.000		
Fisher's Exact Test				.000	.000
Linear-by-Linear Association	88.983	1	.000		
N of Valid Cases[b]	1470				

a. 0 cells (.0%) have expected count less than 5. The minimum expected count is 67.07.

b. Computed only for a 2x2 table

Symmetric Measures

		Value	Approx. Sig.
Nominal by Nominal	Phi	.246	.000
	Cramer's V	.246	.000
N of Valid Cases		1470	

Figure 16: *Crosstab. overtime by attrition*

Then I looked at how the numeric variables relate to each other and to Attrition.

Pearson Correlations

	Attrition	DistanceFromHome	MonthlyIncome	NumCompaniesWorked	PercentSalaryHike	TotalWorkingYears	TrainingTimesLastYear	YearsAtCompany	YearsInCurrentRole	YearsSinceLastPromotion	YearsWithCurrManager
Attrition	1	0.079	-0.198	0.031	-0.024	-0.199	-0.052	-0.19	-0.181	-0.053	-0.175
DistanceFromHome	0.079	1	-0.017	-0.029	0.04	0.005	-0.037	0.01	0.019	0.01	0.014
MonthlyIncome	-0.198	-0.017	1	0.15	-0.027	0.773	-0.022	0.514	0.364	0.345	0.344
NumCompaniesWorked	0.031	-0.029	0.15	1	-0.01	0.238	-0.066	-0.118	-0.091	-0.037	-0.11
PercentSalaryHike	-0.024	0.04	-0.027	-0.01	1	-0.021	-0.005	-0.036	-0.002	-0.022	-0.012
TotalWorkingYears	-0.199	0.005	0.773	0.238	-0.021	1	-0.036	0.628	0.46	0.405	0.459
TrainingTimesLastYear	-0.052	-0.037	-0.022	-0.066	-0.005	-0.036	1	0.004	-0.006	-0.002	-0.004
YearsAtCompany	-0.19	0.01	0.514	-0.118	-0.036	0.628	0.004	1	0.759	0.618	0.769
YearsInCurrentRole	-0.181	0.019	0.364	-0.091	-0.002	0.46	-0.006	0.759	1	0.548	0.714
YearsSinceLastPromotion	-0.053	0.01	0.345	-0.037	-0.022	0.405	-0.002	0.618	0.548	1	0.51
YearsWithCurrManager	-0.175	0.014	0.344	-0.11	-0.012	0.459	-0.004	0.769	0.714	0.51	1

Figure 17: *Pearson correlations: scale variables and Attrition*

Spearman Correlation

	DistanceFromHome	MonthlyIncome	NumCompaniesWorked	PercentSalaryHike	TotalWorkingYears	TrainingTimesLastYear	YearsAtCompany	YearsInCurrentRole	YearsSinceLastPromotion	YearsWithCurrManager
DistanceFromHome	1	0.003	-0.01	0.03	-0.003	-0.025	0.011	0.014	-0.005	0.004
MonthlyIncome	0.003	1	0.19	-0.034	0.71	-0.035	0.464	0.395	0.265	0.365
NumCompaniesWorked	-0.01	0.19	1	0	0.315	-0.047	-0.171	-0.128	-0.067	-0.144
PercentSalaryHike	0.03	-0.034	0	1	-0.026	-0.004	-0.054	-0.026	-0.055	-0.026
TotalWorkingYears	-0.003	0.71	0.315	-0.026	1	-0.014	0.594	0.493	0.335	0.495
TrainingTimesLastYear	-0.025	-0.035	-0.047	-0.004	-0.014	1	0.001	0.005	0.01	-0.012
YearsAtCompany	0.011	0.464	-0.171	-0.054	0.594	0.001	1	0.854	0.52	0.843
YearsInCurrentRole	0.014	0.395	-0.128	-0.026	0.493	0.005	0.854	1	0.506	0.725
YearsSinceLastPromotion	-0.005	0.265	-0.067	-0.055	0.335	0.01	0.52	0.506	1	0.467
YearsWithCurrManager	0.004	0.365	-0.144	-0.026	0.495	-0.012	0.843	0.725	0.467	1

Figure 18: *Spearman correlations: scale variables*

What the matrices told me:

- Nothing numeric is strongly linked to Attrition. The biggest correlations are about −0.20 (TotalWorkingYears, MonthlyIncome, YearsAtCompany).
- Many variables are almost copies of each other: JobLevel and MonthlyIncome (0.92), and the three tenure variables (0.72 to 0.85). The final model uses JobLevel and YearsWithCurrManager but not MonthlyIncome, TotalWorkingYears or YearsAtCompany, which overlap with them.
- Pearson and Spearman agree for most pairs, but not for the tenure variables (0.76 vs 0.85), which means the relationship is steady but not a straight line.
- The satisfaction variables are almost unrelated to each other (|r| < 0.06), so each carries its own information.

Chapter 3: Can I trust my tools?

I also had a question about money: do leavers and stayers earn different amounts once job level is taken into account? To test it I planned a two-way ANOVA (monthly income by attrition and job level). But a test is only trustworthy when its assumptions hold, and ANOVA assumes that the model's *residuals* are roughly normal with similar spread across groups. So before using it, I checked. Raw monthly income is clearly not normal: it has a floor and a cap, so its Q-Q plot bends into an S-shape. But the *residuals* of my ANOVA model were close to normal, because job level explains most of the income pattern.

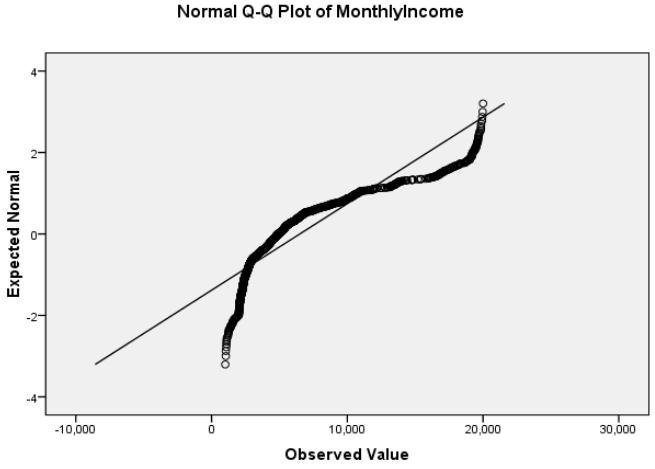


Figure 20: *Q-Q plot of raw MonthlyIncome*

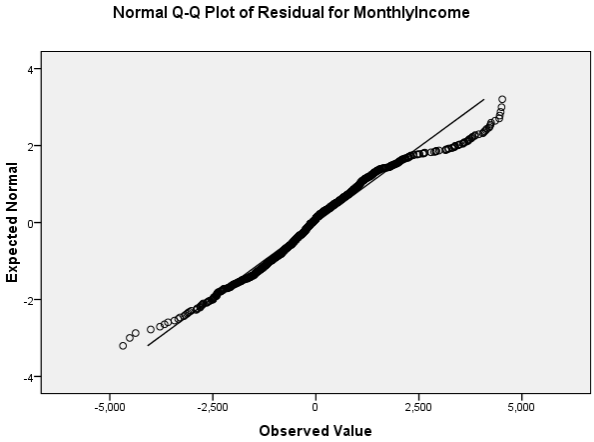


Figure 21: *Q-Q plot of the ANOVA residuals*

The histogram agrees with the Q-Q plot: bell-shaped, centred on 0, with a slightly heavier right tail.

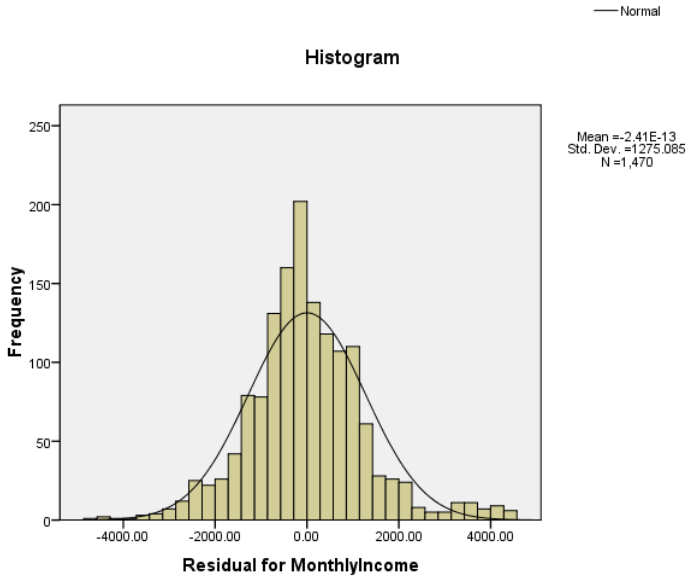


Figure 22: *Histogram of the ANOVA residuals*

The variance was not equal across job levels (narrow at the two ends, wide in the middle), which I learned to read from the residual plot.

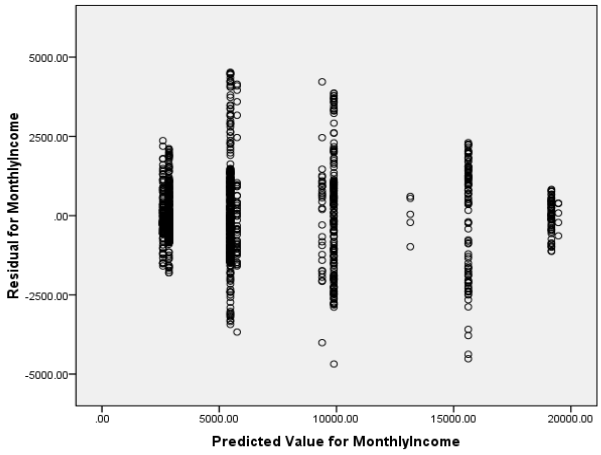


Figure 23: *Residuals vs predicted values of MonthlyIncome*

Then the test itself. **Job level explains most of the income** (partial $\eta^2 = .81$), while **attrition alone explains almost nothing** (partial $\eta^2 = .006$). There is a small interaction (partial $\eta^2 = .016$): leavers and stayers earn about the same at levels 1, 2, 3 and 5, and the only noticeable gap is at level 4, where leavers earn about 2,470 less. Levels 4 and 5 have only 5 leavers each, so I treat that gap carefully. Levene's test showed unequal variances, so I confirmed the interaction with robust (HC3) standard errors: the level-4 coefficient is 2,763 (95% CI 1,912 to 3,614, $p < .001$). In short, money follows job level, which fits the 0.92 correlation between the two and explains why I did not need income as a separate predictor.

Levene's Test of Equality of Error Variances[a,b]

		Levene Statistic	df1	df2	Sig.
MonthlyIncome	Based on Mean	36.617	9	1460	<.001
	Based on Median	30.466	9	1460	<.001
	Based on Median and with adjusted df	30.466	9	1065.118	<.001
	Based on trimmed mean	35.794	9	1460	<.001

Tests the null hypothesis that the error variance of the dependent variable is equal across groups.

a. Dependent variable: MonthlyIncome

b. Design: Intercept + Attrition + JobLevel + Attrition * JobLevel

Tests of Between-Subjects Effects

Dependent Variable: MonthlyIncome

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	3.017E+10[a]	9	3352423603	2049.328	.000	.927
Intercept	2.168E+10	1	2.168E+10	13251.015	.000	.901
Attrition	14248857.20	1	14248857.20	8.710	.003	.006
JobLevel	1.007E+10	4	2516582986	1538.381	.000	.808
Attrition * JobLevel	39562989.72	4	9890747.429	6.046	<.001	.016
Error	2388362609	1460	1635864.801			
Total	9.472E+10	1470				
Corrected Total	3.256E+10	1469				

a. R Squared = .927 (Adjusted R Squared = .926)

Figure 24: *Two-way ANOVA: Levene's test and tests of between-subjects effects*

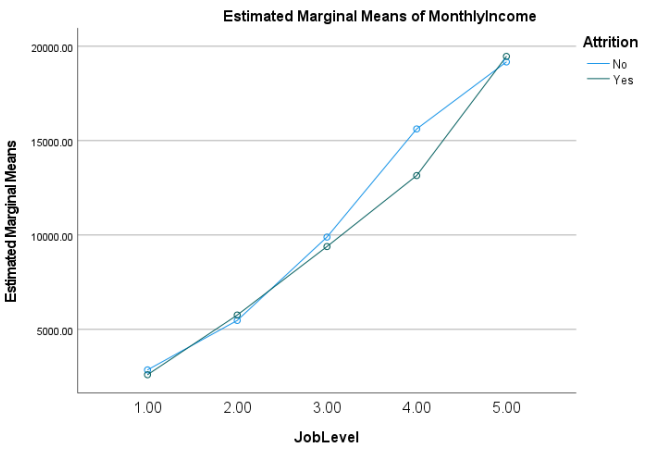


Figure 25: *Profile plot: estimated mean income by job level and attrition*

Parameter Estimates with Robust Standard Errors

Dependent Variable: MonthlyIncome

Parameter	B	Robust Std. Error[a]	t	Sig.	95% Confidence Interval Lower Bound	95% Confidence Interval Upper Bound	Partial Eta Squared
Intercept	19463.800	218.642	89.021	.000	19034.914	19892.686	.844
[Attrition=.00]	-293.222	228.055	-1.286	.199	-740.573	154.129	.001
[Attrition=1.00]	0[b]	.	.	.	.	.	.
[JobLevel=1.00]	-16865.674	227.358	-74.181	.000	-17311.656	-16419.692	.790
[JobLevel=2.00]	-13704.012	310.776	-44.096	<.001	-14313.626	-13094.397	.571
[JobLevel=3.00]	-10075.394	363.336	-27.730	<.001	-10788.109	-9362.678	.345
[JobLevel=4.00]	-6313.400	390.645	-16.161	<.001	-7079.685	-5547.115	.152
[JobLevel=5.00]	0[b]	.	.	.	.	.	.
[Attrition=.00] * [JobLevel=1.00]	549.504	239.308	2.296	.022	80.078	1018.929	.004
[Attrition=.00] * [JobLevel=2.00]	7.929	323.731	.024	.980	-627.098	642.957	.000
[Attrition=.00] * [JobLevel=3.00]	795.848	392.851	2.026	.043	25.236	1566.460	.003
[Attrition=.00] * [JobLevel=4.00]	2763.109	434.017	6.366	<.001	1911.746	3614.472	.027
[Attrition=.00] * [JobLevel=5.00]	0[b]	.	.	.	.	.	.
[Attrition=1.00] * [JobLevel=1.00]	0[b]	.	.	.	.	.	.
[Attrition=1.00] * [JobLevel=2.00]	0[b]	.	.	.	.	.	.
[Attrition=1.00] * [JobLevel=3.00]	0[b]	.	.	.	.	.	.
[Attrition=1.00] * [JobLevel=4.00]	0[b]	.	.	.	.	.	.
[Attrition=1.00] * [JobLevel=5.00]	0[b]	.	.	.	.	.	.

a. HC3 method

b. This parameter is set to zero because it is redundant.

Figure 26: *Two-way ANOVA: parameter estimates with robust (HC3) standard errors*

Chapter 4: Putting everything together

One variable at a time cannot tell me what still matters when the others are held fixed. For that I used logistic regression.

First, I prepared the data carefully:

1. **Overlap check (VIF):** all values below 1.5.
2. **Straight-line check (Box-Tidwell):** the satisfaction variables were not linear, so I treated them as categories.
3. **Merging thin categories:** I merged levels with similar odds ratios or too few people (for example JobLevel 4 with 5, EnvironmentSatisfaction levels 2 with 3, and JobSatisfaction levels 1 to 3 against 4).
4. **Dropping weak variables:** Age and RelationshipSatisfaction.

This took the model from 30 parameters to **22**, and its predictive power barely changed (about 0.01 in AUC), so the simpler model won. I split the employees into **70% for training** (1,060 people, 172 leavers) and **30% for testing** (410 people, 65 leavers).

Here is what the final model found (training set):

Predictor	Odds ratio	95% CI
OverTime: yes vs no	4.99	3.31 to 7.53
YearsWithCurrManager (per year)	0.90	0.83 to 0.97
YearsSinceLastPromotion (per year)	1.09	1.01 to 1.19
BusinessTravel: Travel_Rarely vs Travel_Frequently	0.54	0.34 to 0.87
BusinessTravel: Non-Travel vs Travel_Frequently	0.22	0.09 to 0.53
JobInvolvement: level 2 vs level 1	0.44	0.20 to 0.95
JobInvolvement: level 3 vs level 1	0.27	0.13 to 0.55
JobInvolvement: level 4 vs level 1	0.15	0.05 to 0.41
EnvironmentSatisfaction: levels 2 and 3 (merged) vs level 1	0.32	0.20 to 0.52
EnvironmentSatisfaction: level 4 vs level 1	0.22	0.13 to 0.39
JobSat_High (level 4 vs levels 1 to 3)	0.49	0.32 to 0.77
StockOptionLevel: levels 1 and 2 (merged) vs level 0	0.28	0.18 to 0.43
StockOptionLevel: level 3 vs level 0	0.55	0.26 to 1.18 (not significant)
JobLevel: level 2 vs level 1	0.18	0.08 to 0.43
JobLevel: level 3 vs level 1	0.37	0.13 to 1.03 (not significant)
JobLevel: levels 4 and 5 (merged) vs level 1	0.24	0.05 to 1.07 (not significant)
JobRole: HR / Sales Representative (merged) vs Sales Executive	0.63	0.24 to 1.66 (not significant)
JobRole: Laboratory Technician vs Sales Executive	0.45	0.18 to 1.17 (not significant)
JobRole: Manager / Director (merged) vs Sales Executive	0.11	0.03 to 0.41
JobRole: Manufacturing Director vs Sales Executive	0.31	0.13 to 0.79
JobRole: Research Scientist vs Sales Executive	0.17	0.06 to 0.46
JobRole: Healthcare Representative vs Sales Executive	0.17	0.06 to 0.52

Reference categories: JobRole = Sales Executive, BusinessTravel = Travel_Frequently, StockOptionLevel = 0, JobLevel = 1, JobInvolvement = 1, EnvironmentSatisfaction = 1. JobRole_Final is significant overall (Wald = 38.5, $p < .001$). Compared with Sales Executives, the odds of leaving are significantly lower for Manager/Director, Research Scientist, Healthcare Representative and Manufacturing Director. The merged HR/Sales Representative group and Laboratory Technicians do not differ significantly from Sales Executives once the other variables are held fixed.

Overtime came out as the strongest predictor, in line with what the first chart showed. One thing took me a while to understand: an odds ratio of about 5 does **not** mean overtime workers are 5 times as likely to leave. Odds and probability are different things, so it means the *odds* of leaving are about 5 times higher, with the other variables held fixed.

Omnibus Tests of Model Coefficients

		Chi-square	df	Sig.
Step 1	Step	276.871	22	.000
	Block	276.871	22	.000
	Model	276.871	22	.000

Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	663.148[a]	.230	.391

a. Estimation terminated at iteration number 6 because parameter estimates changed by less than .001.

Hosmer and Lemeshow Test

Step	Chi-square	df	Sig.
1	29.742	8	.000

Contingency Table for Hosmer and Lemeshow Test

		Attrition = No		Attrition = Yes		
		Observed	Expected	Observed	Expected	Total
Step 1	1	105	105.524	1	.476	106
	2	99	104.626	7	1.374	106
	3	104	103.412	2	2.588	106
	4	100	100.724	5	4.276	105
	5	100	99.446	6	6.554	106
	6	99	96.201	7	9.799	106
	7	92	90.556	14	15.444	106
	8	89	82.380	17	23.620	106
	9	69	67.235	37	38.765	106
	10	31	37.896	76	69.104	107

Classification Table[c]

			Predicted					
			Selected Cases[a]			Unselected Cases[b]		
			Attrition			Attrition		
Observed			No	Yes	Percentage Correct	No	Yes	Percentage Correct
Step 1	Attrition	No	866	22	97.5	333	12	96.5
		Yes	104	68	39.5	48	17	26.2
	Overall Percentage				88.1			85.4

Figure 27: *Final model: omnibus test, model summary, Hosmer-Lemeshow and classification at cutoff 0.50*

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)	95.0% C.I.for EXP(B) Lower	Upper
Step 1[a]	OverTime	1.607	.210	58.612	1	.000	4.987	3.305	7.525
	BusinessTravelValue			13.122	2	.001			
	BusinessTravelValue(1)	-.610	.238	6.576	1	.010	.543	.341	.866
	BusinessTravelValue(2)	-1.520	.452	11.334	1	.001	.219	.090	.530
	JobLevel_Final			17.397	3	.001			
	JobLevel_Final(1)	-1.704	.441	14.932	1	.000	.182	.077	.432
	JobLevel_Final(2)	-1.001	.526	3.620	1	.057	.368	.131	1.031
	JobLevel_Final(3)	-1.450	.774	3.509	1	.061	.235	.052	1.069
	JobRole_Final			38.547	6	.000			
	JobRole_Final(1)	-.460	.494	.867	1	.352	.631	.240	1.662
	JobRole_Final(2)	-.792	.483	2.686	1	.101	.453	.176	1.168
	JobRole_Final(3)	-2.216	.678	10.691	1	.001	.109	.029	.412
	JobRole_Final(4)	-1.159	.470	6.081	1	.014	.314	.125	.788
	JobRole_Final(5)	-1.761	.505	12.148	1	.000	.172	.064	.463
	JobRole_Final(6)	-1.783	.572	9.701	1	.002	.168	.055	.516
	EnvironmentSatisfaction_Final			31.652	2	.000			
	EnvironmentSatisfaction_Final(1)	-1.126	.243	21.447	1	.000	.324	.201	.522
	EnvironmentSatisfaction_Final(2)	-1.511	.288	27.429	1	.000	.221	.125	.389
	StockOptionLevel_Final			34.875	2	.000			
	StockOptionLevel_Final(1)	-1.278	.217	34.795	1	.000	.279	.182	.426
	StockOptionLevel_Final(2)	-.591	.386	2.337	1	.126	.554	.260	1.181
	JobSat_Final	-.706	.229	9.534	1	.002	.494	.316	.773
	YearsSinceLastPromotion	.090	.042	4.541	1	.033	1.094	1.007	1.188
	YearsWithCurrManager	-.106	.039	7.362	1	.007	.899	.833	.971
	JobInvolvement			19.389	3	.000			
	JobInvolvement(1)	-.824	.392	4.414	1	.036	.439	.203	.946
	JobInvolvement(2)	-1.325	.371	12.748	1	.000	.266	.128	.550
	JobInvolvement(3)	-1.927	.531	13.172	1	.000	.146	.051	.412
	Constant	3.033	.682	19.768	1	.000	20.766		

a. Variable(s) entered on step 1: OverTime, BusinessTravelValue, JobLevel_Final, JobRole_Final, EnvironmentSatisfaction_Final, StockOptionLevel_Final, JobSat_Final, YearsSinceLastPromotion, YearsWithCurrManager, JobInvolvement.

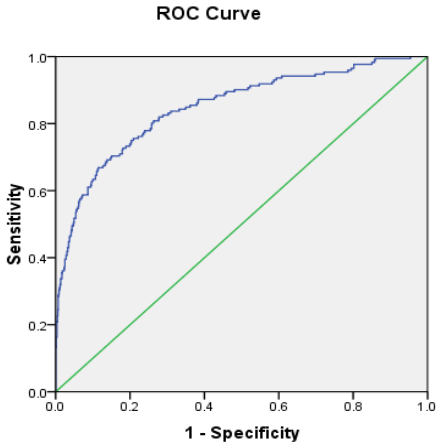
Figure 28: *Final model: variables in the equation (SPSS)*

Case Processing Summary

Attrition	Valid N (listwise)
Positive[a]	172
Negative	888

Larger values of the test result variable(s) indicate stronger evidence for a positive actual state.

a. The positive actual state is Yes.



Diagonal segments are produced by ties.

Area Under the Curve

Test Result Variable(s):Predicted probability

Area	Std. Error[a]	Asymptotic Sig.[b]	Asymptotic 95% Confidence Interval Lower Bound	Upper Bound
.848	.018	.000	.814	.883

Figure 29: *ROC curve, training set (AUC 0.848)*

Chapter 5: The plot twist

At this point I got a test AUC of 0.900 and felt proud. Then I looked at the test output again. It had its own coefficients and its own model fit, and I realized I had refitted the model on the test employees instead of scoring them with the model trained on the 70%. The 0.900 only showed how well a model fits the data it was built on, which is not a test.

I redid it properly: fit on the training set only, then score the test employees. The honest numbers:

	Training	Test
AUC	0.848 (95% CI 0.814 to 0.883)	0.846 (95% CI 0.793 to 0.900)

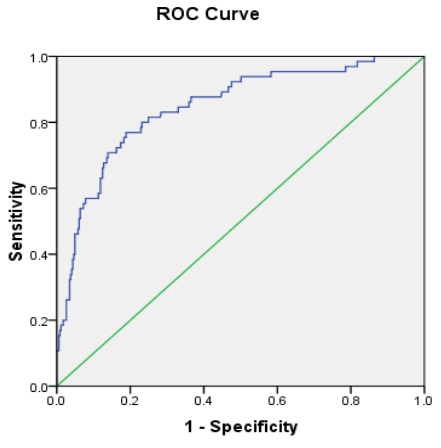
The test AUC is lower than my first result, but it is real, and it almost matches the training AUC, so the model is not just memorizing. An AUC of 0.85 means that if I pick one person who left and one who stayed, the model gives the leaver the higher risk score about 85% of the time.

Case Processing Summary

Attrition	Valid N (listwise)
Positive[a]	65
Negative	345

Larger values of the test result variable(s) indicate stronger evidence for a positive actual state.

a. The positive actual state is Yes.



Diagonal segments are produced by ties.

Area Under the Curve

Test Result Variable(s):Predicted probability

Area	Std. Error[a]	Asymptotic Sig.[b]	Asymptotic 95% Confidence Interval Lower Bound	Upper Bound
.846	.027	.000	.793	.900

Figure 30: *ROC curve, test set (AUC 0.846)*

Chapter 6: From a score to decision

A model gives each person a risk probability, but someone has to decide who to flag. At the default cutoff of 0.50 the model caught only 26.2% of the leavers on the test set, because with only 16% leavers few people reach 50% risk.

So I chose a better cutoff using **the training set only** (Youden's J, which balances catching leavers and avoiding false alarms equally): **0.268**. Then I checked it on the test set:

At cutoff 0.268	Training	Test
Sensitivity (leavers caught)	66.9% (115/172)	58.5% (38/65), 95% CI about 46% to 70%
Specificity (stayers correctly cleared)	88.3% (784/888)	88.1% (304/345)
Precision (flagged people who really leave)	52.5%	48.1% (38/79), about 3 times the base rate of 15.9%

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Attrition * pred	1060	100.0%	0	.0%	1060	100.0%

Attrition * pred Crosstabulation

			pred		
			0	1	Total
Attrition	No	Count	784	104	888
		% within Attrition	88.3%	11.7%	100.0%
		% within pred	93.2%	47.5%	83.8%
	Yes	Count	57	115	172
		% within Attrition	33.1%	66.9%	100.0%
		% within pred	6.8%	52.5%	16.2%
Total		Count	841	219	1060
		% within Attrition	79.3%	20.7%	100.0%
		% within pred	100.0%	100.0%	100.0%

Figure 31: *Crosstab at cutoff 0.268, training set*

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Attrition * pred	410	100.0%	0	.0%	410	100.0%

Attrition * pred Crosstabulation

			pred		
			0	1	Total
Attrition	No	Count	304	41	345
		% within Attrition	88.1%	11.9%	100.0%
		% within pred	91.8%	51.9%	84.1%
	Yes	Count	27	38	65
		% within Attrition	41.5%	58.5%	100.0%
		% within pred	8.2%	48.1%	15.9%
Total		Count	331	79	410
		% within Attrition	80.7%	19.3%	100.0%
		% within pred	100.0%	100.0%	100.0%

Figure 32: *Crosstab at cutoff 0.268, test set*

In plain words, on new employees the model flags about 1 in 5 people and catches about 6 in 10 of those who leave. Nearly half of the people it flags really leave. Accuracy (83.4%) is slightly below "predict everyone stays" (84.1%), which is why I don't use accuracy for a problem where only 16% of people leave.

Chapter 7: What I am not sure about

I want to be clear about the weak spots:

- **Few leavers per variable:** 172 training leavers for 22 parameters is 7.8 per parameter, below the usual guideline of 10 to 15, so some estimates are imprecise.
- **A small leak:** I chose the variables using all 1,470 rows before splitting, so the test results are slightly optimistic. (The JobRole and JobLevel merges were also decided from a model fitted on all rows; the later merges, the dropped variables and the cutoff used the training set only.)
- **Calibration:** the Hosmer-Lemeshow test fails on the training set ($p < .001$). Most of it comes from one low-risk group with 7 leavers instead of the 1.4 expected, and the top-risk group is slightly under-predicted (76 vs 69.1). The model ranks risk well, but its probabilities are not perfect.
- **A small test set:** with only 65 test leavers, a single split is unstable. The AUC could plausibly be anywhere from 0.79 to 0.90.

What comes next

- 5-fold cross-validation for a more stable estimate.
- A calibration plot, and trying an interaction such as OverTime × JobLevel.
- Reproducing everything in Python, and comparing with a random forest using SHAP.
- A Power BI dashboard, and the same crosstabs in SQL.

What I learned

1. Choose the test by the type of data, and report the effect size with the p-value.
2. A model must be judged on data it has never seen.
3. A simpler model that performs the same is the better model.
4. Being honest about a mistake teaches more than a good-looking number.

I'm learning in public. If you spot a mistake or have advice, I would love to hear it.

Appendix: more SPSS output

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
BusinessTravelValue * Attrition	1470	100.0%	0	.0%	1470	100.0%

BusinessTravelValue * Attrition Crosstabulation

			Attrition		
			No	Yes	Total
BusinessTravelValue	Travel_Frequently	Count	208	69	277
		Expected Count	232.3	44.7	277.0
	Travel_Rarely	Count	887	156	1043
		Expected Count	874.8	168.2	1043.0
	Non-Travel	Count	138	12	150
		Expected Count	125.8	24.2	150.0
Total		Count	1233	237	1470
		Expected Count	1233.0	237.0	1470.0

Chi-Square Tests

	Value	df	Asymp. Sig. (2-sided)
Pearson Chi-Square	24.182[a]	2	.000
Likelihood Ratio	23.760	2	.000
Linear-by-Linear Association	23.696	1	.000
N of Valid Cases	1470		

a. 0 cells (.0%) have expected count less than 5. The minimum expected count is 24.18.

Symmetric Measures

		Value	Approx. Sig.
Nominal by Nominal	Phi	.128	.000
	Cramer's V	.128	.000
N of Valid Cases		1470	

Figure A1: *Crosstab: business travel by attrition*

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Department * Attrition	1470	100.0%	0	.0%	1470	100.0%

Department * Attrition Crosstabulation

			Attrition		
			No	Yes	Total
Department	Sales	Count	354	92	446
		Expected Count	374.1	71.9	446.0
	Research & Development	Count	828	133	961
		Expected Count	806.1	154.9	961.0
	Human Resources	Count	51	12	63
		Expected Count	52.8	10.2	63.0
Total		Count	1233	237	1470
		Expected Count	1233.0	237.0	1470.0

Chi-Square Tests

	Value	df	Asymp. Sig. (2-sided)
Pearson Chi-Square	10.796[a]	2	.005
Likelihood Ratio	10.490	2	.005
Linear-by-Linear Association	6.015	1	.014
N of Valid Cases	1470		

a. 0 cells (.0%) have expected count less than 5. The minimum expected count is 10.16.

Symmetric Measures

		Value	Approx. Sig.
Nominal by Nominal	Phi	.086	.005
	Cramer's V	.086	.005
N of Valid Cases		1470	

Figure A2: *Crosstab: department by attrition*

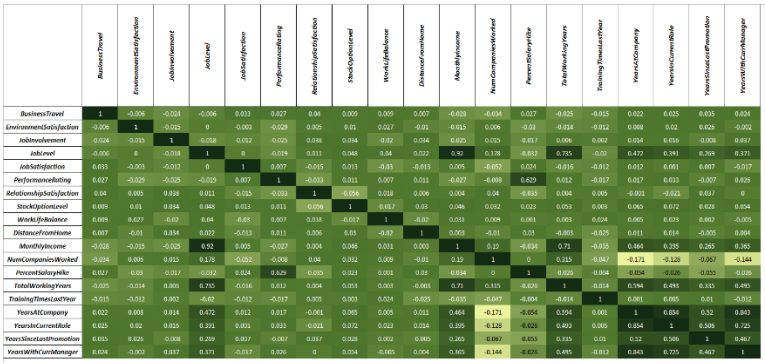


Figure A3: *Spearman correlations: all scale and ordinal variables*

Categorical Variables Codings

			Parameter coding					
		Frequency	(1)	(2)	(3)	(4)	(5)	(6)
Job Role Final Version	Sales Executive	235	.000	.000	.000	.000	.000	.000
	HR/Sales Rep	104	1.000	.000	.000	.000	.000	.000
	Laboratory Technician	187	.000	1.000	.000	.000	.000	.000
	Manager/Director	121	.000	.000	1.000	.000	.000	.000
	Manufacturing Director	103	.000	.000	.000	1.000	.000	.000
	Research Scientist	216	.000	.000	.000	.000	1.000	.000
	Healthcare Representative	94	.000	.000	.000	.000	.000	1.000
Job Level Version 2	1	398	.000	.000	.000			
	2	387	1.000	.000	.000			
	3	156	.000	1.000	.000			
	4	119	.000	.000	1.000			
JobInvolvement	1	59	.000	.000	.000			
	2	264	1.000	.000	.000			
	3	632	.000	1.000	.000			
	4	105	.000	.000	1.000			
Environment Satisfaction Final Version	1	205	.000	.000				
	2	543	1.000	.000				
	4	312	.000	1.000				
Stock Option Level Final Version	0	452	.000	.000				
	1	539	1.000	.000				
	3	69	.000	1.000				
BusinessTravelValue	Travel_Frequently	199	.000	.000				
	Travel_Rarely	745	1.000	.000				
	Non-Travel	116	.000	1.000				

Figure: A4: *Categorical variable codings (training set, n = 1,060). The reference category of each variable is the row where all parameter codings are 0.*

Frequency

JobSat Final Verion

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	0	1011	68.8	68.8	68.8
	1	459	31.2	31.2	100.0
	Total	1470	100.0	100.0	

Stock Option Level Final Version

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	0	631	42.9	42.9	42.9
	1	754	51.3	51.3	94.2
	3	85	5.8	5.8	100.0
	Total	1470	100.0	100.0	

Environment Satisfaction Final Version

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	284	19.3	19.3	19.3
	2	740	50.3	50.3	69.7
	4	446	30.3	30.3	100.0
	Total	1470	100.0	100.0	

Job Level Version 2

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	543	36.9	36.9	36.9
	2	534	36.3	36.3	73.3
	3	218	14.8	14.8	88.1
	4	175	11.9	11.9	100.0
	Total	1470	100.0	100.0	

JobInvolvement

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	83	5.6	5.6	5.6
	2	375	25.5	25.5	31.2
	3	868	59.0	59.0	90.2
	4	144	9.8	9.8	100.0
	Total	1470	100.0	100.0	

Figure A5: *Frequency tables of the final model variables. JobSat_Final: 1 = level 4, 0 = levels 1 to 3. StockOptionLevel: 0, 1 (original 1 and 2), 3(original 3). EnvironmentSatisfaction: 1, 2 (original 2 and 3), 4. JobLevel: 4 = original 4 and 5.*